Abhishek Kumar

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Professional Summary

Quantum-algorithm researcher specializing in early and fully fault-tolerant methods for dynamical correlators and time-dependent systems. Developed the early fault-tolerant algorithm SPICE-Q for multipoint dynamical correlator estimation with precision-independent coherent circuit depth, and the variational algorithm Floquet-ADAPT-VQE for periodically driven systems. Experienced in algorithm design, numerical benchmarking, and logical resource estimation. Broadly interested in quantum algorithms across different fields, their resource estimation, and their implementation on quantum hardware.

Core Expertise

- **Quantum algorithms:** (i) Hamiltonian simulation using product formulas, Taylor-series and LCU methods, and QSVT; (ii) sampling-based early fault-tolerant algorithms; and (iii) variational algorithms, including VQE and ADAPT-VQE
- **Algorithm-level resource estimation:** Logical-qubit, Clifford-gate and T-count estimates, and sample complexity
- **Quantum simulation** for many-body phenomena and many-body time-dependent systems
- **Condensed matter theory** and electronic-structure problems
- **Computational tools:** Qiskit, Python, Julia, C++, Git, and HPC

Professional Experience

Postdoctoral Associate — **Aug 2022–Present**

Department of Physics, Virginia Tech — *Blacksburg, USA*

Advisors: Prof. Edwin Barnes and Prof. Sophia E. Economou

Research Focus:

- Near-term and fault-tolerant quantum algorithms for time-dependent systems
- Early fault-tolerant algorithms for dynamical quantities, including multipoint dynamical correlators and out-of-time-order correlators in quantum many-body systems
- Hybrid CV–DV frameworks for quantum simulation
- Nonergodicity and time crystallinity in central-spin models

Selected Quantum Algorithm Projects

Floquet-ADAPT-VQE

- Developed and numerically validated a hybrid quantum-classical framework for quasienergy estimation and Floquet-state preparation in periodically driven systems.
- It enables fixed-depth access to time-dependent micromotion expectation values in a Floquet state.

SPICE-Q: An Early Fault-Tolerant Algorithm for Estimating Multipoint Dynamical Correlators with Precision-Independent Coherent Circuit Depth [Draft is available upon request, Soon in arXiv]

- Developed SPICE-Q (*Sampling-based Precision-Independent Correlator Estimation on Quantum computers*), a randomized algorithm for estimating general multipoint dynamical correlators.
- Designed a circuit construction requiring only one ancilla, with per-sample coherent gate complexity independent of the target precision.
- Derived rigorous correlator-approximation and statistical-sampling error bounds and analyzed logical-qubit, CNOT, rotation, and T-gate requirements.
- Benchmarked the method on quantum many-body and molecular Hamiltonians relevant to quantum transport, information scrambling, and quantum chaos.

Novel Taylor-Series Expansion for the Time-Evolution Operator of Smooth Time-Dependent Hamiltonians [Ongoing project]

- Derived a novel Taylor-series expansion for time evolution generated by smooth time-dependent Hamiltonians, with efficiently computable coefficients and without explicit time-ordered integrals.
- The framework is advantageous over Dyson-series methods for time-periodic Hamiltonians with only a few Fourier modes in LCU-based implementations.

Education

Ph.D. in Physics **2016–2022**
Indiana University *Bloomington, USA*

Thesis: *Study of Dynamics and Topology of Driven Many-Body Quantum Systems*
Advisor: Prof. Babak Seradjeh

Integrated M.Sc. in Physics **2011–2016**
National Institute of Science Education and Research (NISER) *Bhubaneswar, India*

Thesis: *Study of Weyl-Semimetal–Superconductor Junctions*

Publications

- "Floquet-ADAPT-VQE: Quantum Algorithm to Simulate Nonequilibrium Physics in Periodically Driven Systems," **A. Kumar**, K. Shirali, N. Mayhall, S. E. Economou, and E. Barnes; arXiv:2503.11613 (2025)
- "Hilbert-Space Fragmentation and Subspace-Scar Time Crystallinity in Driven Homogeneous Central-Spin Models," **A. Kumar**, R. Frantzeskakis, and E. Barnes; Phys. Rev. B **111**, 014302 (2025)
- "Enhancement of High Harmonic Generation in Bulk Floquet Systems," **A. Kumar**, Y. Li, B. Seradjeh; arXiv:2207.02830 (2022)
- "Floquet Gauge Pumps as Sensors for Spectral Degeneracies Protected by Symmetry or Topology," **A. Kumar**, G. Ortiz, P. Richerme, and B. Seradjeh; Phys. Rev. Lett. **126**, 206602 (2021)
- "Linear-Response Theory and Optical Conductivity of Floquet Topological Insulators," **A. Kumar**, M. R. Vega, T. P. Barnea, and B. Seradjeh; Phys. Rev. B **101**, 174314 (2020)
- "Higher-Order Floquet Topological Phases with Corner and Bulk Bound States," M.R. Vega, **A. Kumar**, B. Seradjeh; Phys. Rev. B **100**, 085138 (2019), Editors' Suggestion

Awards and Honors

- College of Arts and Sciences Travel Award, Indiana University (2021)
- Indiana University Graduate Fellowship (2016–2017)
- INSPIRE Scholarship, Government of India (2011–2016)
- Visiting Student Program Fellowship, HRI Allahabad (2015)

Leadership, Service, and Outreach

- Referee for *Physical Review Letters*, *Physical Review A*, *Physical Review B*, and Springer Nature journals
- Co-organizer, VTQ Qiskit Hackathon (2025)
- Instructor, Virginia Tech Quantum Summer School (2024, 2025)
- APS Session Chair (2024, 2025) and APS Quantum To-Go outreach speaker (2024)
- Co-organizer, QIS Workshop for Early-Career Researchers (Summer 2023)